

MUNICH STADT, Germany

WMO: 108650

Lat: 48.1632N

Lon: 11.5429E

Elev: 526

StdP: 95.17

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.9	-9.2	-15.1	1.1	-10.4	-12.8	1.3	-7.9	10.3	9.2	9.2	7.1	1.8	90	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	29.5	19.0	27.7	18.1	26.1	17.5	19.7	27.4	18.9	26.1	18.2	24.4	2.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	13.1	21.6	16.4	12.5	21.1	15.8	11.9	20.8	58.6	27.5	55.9	26.3	53.3	24.5	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.8	6.4	5.3	DB	-13.4	33.1	4.3	2.0	-16.5	34.6	-19.0	35.8	-21.4	36.9	-24.5	38.4	(4)
(5)			WB	-13.8	21.2	4.2	0.8	-16.8	21.8	-19.3	22.3	-21.6	22.7	-24.6	23.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	0.5	0.5	5.0	8.5	13.6	16.5	19.5	18.9	15.3	10.3	4.4	1.8
	DBStd	7.94	5.18	5.16	4.23	3.83	3.88	3.87	3.39	3.45	3.30	3.80	3.83	4.33
	HDD10.0	1287	294	268	163	74	14	2	0	0	1	43	172	256
	HDD18.3	3337	552	501	414	296	151	81	27	35	101	248	419	513
	CDD10.0	1148	0	1	7	28	125	197	295	277	161	53	3	1
	CDD18.3	157	0	0	0	0	3	26	64	53	11	0	0	0
	CDH23.3	1184	0	0	0	2	27	186	491	403	72	4	0	0
	CDH26.7	279	0	0	0	0	1	33	132	106	7	0	0	0
(14) Wind	WSAvg	2.1	2.4	2.3	2.5	2.3	2.1	2.1	1.9	1.8	1.8	1.8	2.1	2.1
(15) Precipitation	PrecAvg	981	56	48	64	63	108	127	126	116	80	68	65	61
	PrecMax	1187	141	106	132	139	206	199	268	199	152	132	147	114
	PrecMin	685	2	14	22	15	9	60	34	45	20	14	0	1
	PrecStd	117	31	26	32	35	46	45	47	43	35	30	33	29
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	16.0	20.7	22.1	25.8	29.9	32.9	31.9	27.7	23.6	15.9	14.1
		MCWB	7.6	9.6	11.7	12.4	16.1	19.2	20.5	19.4	18.1	15.2	10.4	8.2
	2%	DB	10.5	12.3	16.2	19.7	23.9	27.3	29.8	29.5	25.3	20.6	13.5	11.8
		MCWB	6.6	7.6	9.1	11.3	15.0	17.7	19.3	18.8	17.1	14.1	9.3	7.6
	5%	DB	8.9	9.5	13.7	17.5	22.1	25.6	27.8	27.5	23.5	17.9	11.6	9.7
		MCWB	5.6	5.9	8.0	10.5	13.9	17.1	18.3	18.0	16.4	12.9	8.5	6.6
	10%	DB	7.3	7.2	11.3	15.3	20.4	23.5	26.1	25.5	21.5	16.0	9.9	7.9
		MCWB	4.7	4.7	7.0	9.5	13.2	16.0	17.9	17.3	15.6	12.2	7.6	5.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	9.8	11.8	13.1	17.6	19.9	21.3	20.3	19.0	16.2	11.3	9.7
		MCDB	11.2	15.2	20.4	20.0	23.1	27.4	29.9	29.0	25.3	21.7	14.1	12.6
	2%	WB	7.3	8.0	9.9	12.0	15.7	18.7	20.2	19.5	17.9	14.9	9.9	8.2
		MCDB	10.4	11.9	14.8	18.5	21.3	25.4	28.0	27.4	23.6	18.8	12.4	10.8
	5%	WB	5.9	6.3	8.5	11.1	14.7	17.7	19.2	18.7	17.1	13.5	8.8	7.0
		MCDB	8.2	8.9	12.4	16.3	20.4	23.8	26.3	25.8	22.2	17.2	11.2	9.3
	10%	WB	4.7	4.9	7.6	10.0	13.9	16.7	18.3	17.9	16.1	12.4	7.6	5.7
		MCDB	6.9	7.2	11.0	14.6	19.2	22.4	24.5	23.9	20.5	15.4	9.7	7.9
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.2	7.1	7.9	8.5	8.3	8.8	8.9	8.1	7.0	5.8	5.0
		MCDBR	7.5	10.3	11.3	12.3	11.7	11.6	12.1	12.4	11.7	11.2	9.2	8.0
	5% WB	MCWBR	4.9	6.2	5.7	5.9	5.1	4.3	4.6	4.1	5.0	5.7	5.6	5.2
		MCWBR	6.6	9.4	9.8	11.0	10.2	10.5	10.9	10.9	10.0	9.3	8.3	7.1
(40) Clear-Sky Solar Irradiance	taub		0.296	0.321	0.354	0.389	0.398	0.403	0.406	0.400	0.376	0.346	0.318	0.292
		taud	2.448	2.384	2.342	2.275	2.283	2.305	2.289	2.325	2.370	2.447	2.481	2.468
	Ebn at Noon		783	836	859	861	864	861	851	840	826	796	749	742
		Edh at Noon	64	85	104	123	127	125	126	116	101	80	63	56
(44) All-Sky Solar Radiation	RadAvg		1.09	1.97	3.09	4.29	4.95	5.54	5.36	4.71	3.41	2.18	1.18	0.88
	RadStd		0.14	0.26	0.38	0.57	0.57	0.50	0.49	0.48	0.41	0.26	0.16	0.12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (91 neighbors)	+0.37	N/A	N/A	N/A	+0.33	+0.34	N/A	N/A	+55	+19

Nomenclature: See separate page